TP 2 Supervised Learning 2A
2025-2026

Duration 3h
Use of ChatGPT / Claude / other LLM encouraged.

This session is dedicated to the study of a single dataset, on which you will implement all the discrimination models covered in class so far (nearest neighbors algorithm, naive Bayes, discriminant analysis).

Part 1: Loading the dataset, exploratory analysis

This dataset consists of measurements taken on several wines. For each wine sample, 11 characteristics were measured (acidity, density, pH, etc.) and after tasting, each received a score. For the purposes of this assignment, we will consider that each wine was assigned a class 1 (good quality wine) or class 0 (poor quality wine). The goal is to predict the class of a wine based on its 11 characteristics.

Questions:

1. Load the dataset and analyze the summary of the table. What do you notice?

2. Using a preliminary visual exploration of the data, identify the characteristics that appear at first glance to most influence the quality of a wine.

Part 2: Sample construction, model training

Now it's your turn to get familiar with the models covered in class, by following these instructions:

1. Start from the same initial point as in the last exercise of the previous assignment, namely to build a training sample and a test sample. For this step, it is important to take into account what you noticed in question 1 above!

2. The models you need to train are the following
    a. Naive Bayes
    b. Linear and Quadratic Discriminant Analysis
    c. Nearest Neighbors Algorithm

    For each of them, do not forget to optimize their hyperparameters (recall the hyperparameters of each, if applicable, and provide the procedure you should use to optimize them).

3. Compare all the models with each other on the test sample and identify the most predictive one.